MCS-203	MOBILE COMPUTING	L T P 3 1 0	4 Credits

Course Objectives:

CO1	To learn about the basic concepts of Mobile Computing.
CO2	To understand about networking concepts relevant to modern wireless systems.
CO3	To introduce emerging mobile computing ideas and best practices
CO4	To gain hands-on knowledge practice with mobile computing
CO5	To understand various types of Protocols

MODULE I

Issues in Mobile Computing, Overview of wireless Telephony, IEEE 802.11 & Blue Tooth, Wireless Multiple access protocols, channel Allocation in cellular systems. Data Management Issues, data replication for mobile computers, adaptive Clustering for Mobile Wireless networks.

MODULE II

Distributed location Management, pointer forwarding strategies, Energy Efficient Indexing on air, Energy Indexing for wireless broadcast data, Mobile IP, TCP Over wireless.

Mobile Agents Computing, Security and fault tolerance, transaction processing in Mobile computing environment.

MODULE III

Ad hoc network, Routing Protocol, Introduction – Issues in Designing a Routing Protocol for Ad Hoc Wireless Networks – Classifications of Routing Protocols – Table–Driven Routing Protocols – Destination Sequenced Distance Vector (DSDV) – Wireless Routing Protocol (WRP) – Cluster Switch Gateway Routing (CSGR) – Source–Initiated On–Demand Approaches – Ad hoc On–Demand Distance Vector Routing (AODV) – Dynamic Source Routing (DSR) –Temporally Ordered Routing Algorithm (TORA) – Signal Stability Routing (SSR) –Location–Aided Routing (LAR) – Power–Aware Routing (PAR) – Zone Routing Protocol (ZRP). Itroduction and application of Vehicular Communication.

References

1. *J. Schiller, "Mobile Communications", Addison Wesley.*

2. *A. Mehrotra, "GSM System Engineering : Mobile Communication Series", Artech House Publishers, ISBN: 0890068607.*
3. *M. V. D. Heijden, M. Taylor, "Understanding WAP", Artech House.*
4. *Charles Perkins, "Mobile IP", Addison Wesley.*
5. *Charles Perkins, "Ad hoc Networks", Addison Wesley.*

Course Outcome:

CO1	Various wireless communication technologies.
CO2	Enables the students to visualize the various important steps in GSM communication
CO3	To acquire the knowledge of the fuzzy Neural network and Genetic Language
CO4	Enables the students to analyze the mobile IP and Transport Protocol.
CO5	To analyze various type of Protocols